

AI acknowledgement coversheet

This coversheet is designed to help you acknowledge any AI use in your assessment. If your course coordinator provides a course-specific template, please use that instead. The AI Student Hub has more information about using AI in assessment.

Your details

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Assessment details

Course: DECO7230 Digital Prototyping and Extended Reality

Assignment: Interactive Prototype Outcome Evaluation 2 (Submission 3)

AI use

Select the option that applies:

- ☒ Yes, AI tools have been used to generate or edit material in this document
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Details of use

Record the following information when acknowledging the use of AI tools:

- AI tool (e.g. Copilot, ChatGPT, Claude, Gemini)
- description of how you used the tool (e.g. edited, corrected, translated, planned, brainstormed)
- prompt(s) used
- relevant part, section, or page of the assessment
- date.

Complete the table below with this information.

Tool	Use	Prompt(s)	Section	Date
Claude Fable 5.1 (via Claude Code)	Exported my five iPad logger records and generated the results table straight from the JSON (no hand transcription), with raw counts per assumption and a data-quality list; I checked it against the logger on the day	Export the iPad logger records to testing-data/ip2a and generate the results table, with the counts for each assumption.	Section 2 (Results), Appendices A–D	25 September 2026
Claude Opus 5.5 (via Claude Code)	Drafted the analysis from my results and testing plan: patterns, verdicts on A1’–A4, and the data caveats (e.g. prompt timing). It also checked the colour generator code behind insight 3. I reviewed every verdict against the data	What do we need to do for the submission in two days? Then: Yes, start drafting Evaluation 2.	Sections 3–4, Appendix C	5 October 2026
Claude Opus 5.5 (via Claude Code)	Proposed the concept iterations and reflection points (fan-deck sample, lightness-based palette, toss-to-commit as a test candidate, real light in MR,	Yes, start drafting Evaluation 2.	Sections 5–6	5 October 2026

Tool	Use	Prompt(s)	Section	Date
	asking A2 by exposure). I decide which go into IP2b			
Claude Opus 5.5 (via Claude Code)	Generated the first draft of the whole document and its print layout (HTML/CSS to PDF) to the six-section template, 3 pages plus appendices, then reformatted it at my direction for clarity and bigger headings; I own every judgement in it	Yes, start drafting Evaluation 2. Then: The PDF is confusing; can you make the format clearer? Then: Can you make the section titles larger so they stand out?	Entire document	5 October 2026

Prompts are lightly edited for spelling and grammar; their meaning is unchanged. The prototype, testing plan, iPad session logger and on-device setup were also built with AI assistance; that use is logged in `ip2a/ai-use-log.md` in the repository. The test sessions and all data entry were done by me.

This coversheet has been adapted with thanks to BIOM3020 Integrated Endocrinology.

XR Renovation Previewer

Evaluation of Interactive Prototype 2a · Week 9 studio, Fri 25 Sep 2026

Kaike Nehme · s4959287 · Semester 2, 2026 · github.com/kaikenehme/deco7230-xr · raw data in Appendices A–D

AT A GLANCE	
Tested	5 participants, all on the Meta Quest 3S headset (the plan said desktop mode; see note in §1).
Fixed	The way into the hero interaction: 4 of 5 pulled a colour sample without help (IP1: 0 of 5). Furniture: 5 of 5 placed it first try.
Found	Everyone still kept a colour from the menu (5 of 5), because it was easier: nobody saw more than 2 of the 7 matching colours the sample offers.
Still open	Whether limiting the options <i>helps</i> (A2). My interview question never reached anyone.
Next	Show all 7 colours at once, build them around lightness, test in the participant's real room (IP2b).

01 Objective and validation metrics

CATEGORY: CREATION

The prototype. A furnished virtual living room. The floor and sofa are **staying**; the walls can change. The hero interaction: touch a kept surface to pull off a colour sample, **hold it up** against a wall to preview a matching paint, twist the wrist to step through 7 matching options, let go to commit. A normal menu with the full paint catalogue is available too, as the comparison. New since IP1: a visible tab and glow when you touch a kept surface, furniture you can grab, place and rotate, a clock that changes the daylight, a lamp, and three preset rooms.

AIM	QUESTION	HOW IT WAS MEASURED
A1' · Getting in	Do people pull a sample without being told , now that touching gives feedback?	Pulled a sample during the silent first 45 s · noticed the tab/glow · found the wrist twist
A2 · Constraint	Does limiting options to matching colours help people decide? (primary aim)	"Helpful or restrictive?" · options tried · confidence 1–5 · which colour was rejected and why
A3 · Route	Hold-up or menu: which do people use for the final colour, and which do they trust ?	Method of the kept colour · "why the menu?" · "which did you trust more?"
A4 · Light	Do people use time of day, and does it change a decision ?	Clock used, stops visited · "did the time of day change your mind?"
Furniture	Do the grab / place / rotate fixes from IP1 work first time?	Armchair placed first try · rotate found · lamp found

Session (~7 min each): brief → 45 s silent free look → Task 1, starting with a floor sample already in hand: *"Repaint this room so it works with the floor and sofa"* → Task 2: move the armchair, add a piece, check another time of day → 6 interview questions, logged live on an iPad. No numeric pass marks were set, so aims are judged on how consistent the evidence is.

Platform change. The testing plan says **desktop mode** (mouse and keyboard) because the headset had failed to connect before IP1. On the day it connected, so all five sessions ran **on the Quest 3S**. The plan is left as written. This means IP2a is real headset data, but any comparison with IP1 (simulator) mixes a design change with a platform change.

02 Results

5 participants, all studio peers. P4 and P5 had seen the project before. Nobody agreed to recording. — = not recorded.

What they did

BEHAVIOUR	P1	P2	P3	P4	P5	TOTAL	IP1
GETTING IN (A1')							
Noticed the tab or glow	✓	✓	✓	✓	✓	5/5	—
Pulled a sample without help	✓	✓	✓	—	✓	4/5	0/5
Found the wrist twist	✗	✗	✓	✓	✓	3/5	0/3
CHOOSING A COLOUR (A2, A3)							
Matching options seen (of 7)	2	2	2	1	2	max 2	—
Kept colour came from	menu	menu	menu	menu	menu	menu 5/5	menu 5/5
Confidence in kept colour (1–5)	5	4	4	4	5	avg 4.4	avg 3.0
LIGHT (A4) AND FURNITURE							
Clock stops visited (of 6)	5	3	4	5	3	all ≥3	—
Armchair placed first try	✓	✓	✓	✓	✓	5/5	—
Found rotate without help	✓	✓	✓	✓	✓	5/5	—
Found the lamp without help	✗	✓	✗	✓	✓	3/5	2/5

What they said

Why the menu?	"Easier to use" (P3) · "More intuitive" (P4) · "Easy setup" (P5)
Trusted more	Menu: "easier to use" (P3), "more intuitive" (P4) · From the room: "misses more physical interaction" (P1)
Rejected colour	"Light colour, personal preference" (P1) · "Light colour, personal preference" (P3) · "Black colour, personal" (P4)
Time of day changed a decision?	"No" (P1, P3) · not asked of the others
Confusing / ideas	"The drag from the furniture" (P1) · "Too sensitive, the furniture moving" (P4) · "Everything was pretty smooth" (P3) · ideas: a physical lamp, "a balloon thrown at the wall changes the colour" (P1), "other types of room" (P2)

Data caveats (full list in Appendix C): prompt timing wasn't captured, so it's unknown whether the hint "press the trigger on the tab" — logged for all four who pulled — came before or after their pull. The hint "touch the clock" was given to all five, so clock use was not discovery.

03 Analysis and insights

1 • The way in now works. Everyone noticed the tab or glow and 4 of 5 pulled a sample; in IP1 the same touch led nowhere for all 5. Caution: the headset also changed, and a hint may have come first for some, so I can't say how much is the new design alone.

2 • People preview with the sample but decide with the menu. All five held a matching sample; all five kept a menu colour, saying it was easier. Why: nobody saw more than 2 of the 7 options, and 2 of 5 never found the twist that shows the rest. It was two hidden colours against a whole visible catalogue. That tests visibility, not whether limiting options helps.

3 • People judged on lightness, the generator varies hue. All three rejections were about lightness ("light", "black"), and nobody mentioned the floor or sofa. The colour generator changes hue in 5 of 7 options and leaves out off-whites, which is what most real walls are.

4 • The menu wins on effort, the room wins on feel. Menu-trusters said "easier" and "intuitive". The one room-truster wanted *more* physical interaction (a physical lamp, throwing colour at the wall). The hold-up route loses for as long as it is also the slower one.

5 • Furniture is solved, it needs tuning. IP1's biggest complaint is gone (5/5 placed, 5/5 rotated). What's left is "too sensitive" and "the drag".

6 • Time of day was used because they were told to. Everyone used the clock, but the task and a hint asked them to. Neither of the 2 asked changed a decision.

04 Evaluation of aims

AIM	VERDICT	WHY	CONFIDENCE
A1' · Getting in	SUPPORTED	4/5 pulled, 5/5 noticed. The twist (3/5) is now the weak step	Medium: mixed with platform change
A2 · Constraint	UNTESTED	The "helpful or restrictive?" question was only asked of people who <i>kept</i> a hold-up colour, and nobody did. My instrument error	Low: behaviour leans against the constraint as built
A3 · Route	ANSWERED	Menu 5/5 for the kept colour, with both routes working. Reason: ease	Medium for use, low for trust (3 answers)
A4 · Light	PARTLY	Used by all, but on instruction; changed 0 of 2 decisions	Low
Furniture	VALIDATED	5/5 placed first try, 5/5 found rotate (IP1: 1/5 grab)	Medium

Summary: IP2a fixed the way in and the furniture, and measured a real preference for the first time: the menu, on ease. The primary aim (A2) is still untested: in IP1 the prototype blocked it, in IP2a my question did, because I asked based on *what people kept* instead of *who held a sample*.

05 Concept iteration

The concept (match against what's staying, hold it up, no menus) is **kept but not yet proven**. Changes are framed for IP2b, which runs in the participant's **real room** (MR passthrough).

BECAUSE...	CHANGE	TYPE
The way in works (insight 1)	Keep the tab, glow and peel; in MR they attach to the participant's own floor and furniture. This also answers "other types of room".	Affirm
Hidden options lose to a visible catalogue (2)	The sample fans open in the hand like a paint fan deck : all 7 colours visible at once, pick one with the other hand, hold it to the wall. Still a held object, not a menu.	Change
People judge on lightness (3)	Build the 7 options light to dark , always including two off-whites. Keep the thread back to the source.	Change
Room wins on feel, wants more physical (4)	Behind "throw a balloon at the wall": committing from a distance, and play. Test tossing the chip at the wall against hold-up. Lamp becomes a pull cord you grab.	Test
Furniture "too sensitive" (5)	Smooth the movement and slow the rotation.	Fix
Time of day changed nothing (6)	In MR the room's real light replaces the clock; the clock stays in VR mode only.	Change
A2 never asked	Ask "helpful or restrictive?" of everyone who held a sample , and add a short phase before the menu appears.	Test plan

06 Reflection and next steps

WHAT WORKED	WHAT DIDN'T	WHAT I'LL CHANGE
- Switching to the headset gave real reach and scale data - The device pipeline now works (72 fps), removing IP2b's biggest risk - The iPad logger lost no rows and logged which hint was used - Starting Task 1 with a sample in hand got everyone to the matching colours	- The platform switch broke comparison with the plan and with IP1 - My interview branch skipped the primary question - Phase timers unused, so hint timing was lost - Studio peers only; 2 had seen the project	- Branch questions on <i>exposure</i>, not outcome - One "next phase" button that timestamps everything - IP2b plan: standardised questionnaire, ≥5 academic references, the five design principles - Build: fan deck, lightness palette, toss-to-commit, real light

The main lesson. Twice now, the primary aim went unanswered for a reason unrelated to the concept: in IP1 people never reached the constrained colours; in IP2a they reached them but were never asked about them. Both times I designed the test around the path I expected people to take. For IP2b, every decisive question is asked of everyone who was *exposed* to the thing it asks about, whatever they chose. The platform switch was the right call for data quality, but I'll decide the platform before writing the plan, not on the day.

Appendices

A full logger results · B prompts as logged · C data-quality and transcription notes · D session and device notes

A Full results from the iPad logger, Fri 25 Sep 2026, Week 9 studio (n = 5, Meta Quest 3S)

Exported the same day from the session logger to `testing-data/ip2a/logger-export/participants/p1-p5.json`. Text is as typed live on the iPad, spelling included. Blank (—) = not recorded. Facilitator: Kaike Nehme, solo. No recordings (consent declined by all five).

#	MEASURE	P1	P2	P3	P4	P5
—	Seen the project before	No	No	No	Yes	Yes
—	Recording consent	No	No	No	No	No
1	Reached for first	The furniture	Walls	Sofa	Furniture	Floor
2	Tab or glow noticed	Yes	Yes	Yes	Yes	Yes
3	Sample pulled in free look	Unprompted	Unprompted	Unprompted	—	Unprompted
4	Prompts used (numbers, logged order)	1, 4, 3, 8	1, 3, 6, 8	3, 5, 3, 2, 8	1, 2, 6, 8	3, 5, 7, 6, 8
5	Time to first unprompted commit	1 min	2 min	3 min	4 min	4 min
6	Twist-to-tune found	No	No	Unprompted	Unprompted	Unprompted
7	Harmonised options tried before first commit	2	2	2	1	2
8	Method for the kept colour	Menu	Menu	Menu	Menu	Menu
9	Thread noticed	Yes	Yes	—	Yes	Yes
10	Armchair placed first try	Yes	Yes	Yes	Yes	Yes
11	Rotate found	Unprompted	Unprompted	Unprompted	Unprompted	Unprompted
12	Clock used · stops visited	Yes · 5	— · 3	Yes · 4	Yes · 5	Yes · 3
13	Lamp found	Prompted	Unprompted	Prompted	Unprompted	Unprompted
14	Menu opened · furniture	Unprompted · Added furniture	Unprompted · Added furniture	Unprompted	Unprompted	Unprompted
15	Preset switched	Yes	Yes	Yes	Yes	Yes
16	Confidence 1–5	5	4	4	4	5
17a	Helpful or restrictive? (hold-up users)	—	—	—	—	—
17b	Why the menu? (menu users)	—	—	Easier to use	More intuitive	Easy setup
18	Rejected colour, why	Light colour , personal preference	—	Light colour by personal preference	Black colour personal	—
19	Trusted more, why	From the room, she misses more physical interaction	—	Menu, easier to use	Menu, more intuitive	—
20	Time of day changed a decision	No	—	No	—	—
21	One thing that confused them	The drag from the furnitures	—	Everything was pretty smooth	—	—

B Prompts as logged, with the scripted wording

PROMPT NUMBERS, SECONDS AFTER THE FIRST PROMPT P LOGGED	# SCRIPTED PROMPT (ONLY AFTER ~20 S STUCK)
P1 1 (0 s) · 4 (5 s) · 3 (102 s) · 8 (111 s)	1 "The floor and sofa are staying — everything else can change."
P2 1 (0 s) · 3 (8 s) · 6 (10 s) · 8 (12 s)	2 "Move your hand close to the floor or the sofa."
P3 3 (0 s) · 5 (73 s) · 3 (138 s) · 2 (139 s) · 8 (158 s)	3 "Press the trigger on the little colour tab."
P4 1 (0 s) · 2 (1 s) · 6 (219 s) · 8 (230 s)	4 "Try holding it near a wall."
P5 3, 5, 7, 6, 8 all within 1.5 s (entered afterwards)	5 "Try twisting your wrist while holding it."
Every prompt is stamped phase "task 1" because the phase timers were not started. Gaps of a second or less mean entry after the fact, not delivery.	6 "Point at the floor and press the trigger." (opens the menu)
	7 "Squeeze the grip to hold the furniture; let go to place it. Push the thumbstick to turn it."
	8 "Touch the clock by the window."

C Data-quality and transcription notes

- **Prompt timing and order are unreliable** (Appendix B). Which prompts were used is reliable; when, and in which phase, is not.
- **Row 3 against prompt 3.** P1, P2, P3 and P5 are logged "unprompted" in row 3, and each also has prompt 3 ("press the trigger on the little colour tab") in their list. Since Task 1 begins with a sample placed in the hand, prompt 3 may have been used later to pull a second sample, but the log cannot show this. A1' is reported as provisional for this reason.
- **Prompt 8 for all five.** Clock use (row 12) followed both the Task 2 instruction and, at some point, prompt 8, so it is not a discovery measure.
- **Prompt 6 for P2, P4, P5** reveals the menu, so for those three the menu was shown at some point although row 14 (Task 2) logs it opened unprompted. P1 and P3 reached the menu without prompt 6. **Prompt 7 for P5** names the thumbstick, although row 11 logs rotate as unprompted.
- **Row 5** was typed in whole minutes, not read off the stopwatch. P4 and P5's 4 min exceed the 2-minute task: either the task ran long or "4" means "not within the task".
- **Row 17a was never asked.** It branched on row 8 = hold-up or both; row 8 was menu for all five. This is an instrument design error, not a missing entry.
- **Blanks are blanks**, not "no": P4 row 3, P3 row 9, P2 row 12 (yes/no), and several of rows 17–21.
- The logger's prompt wording had been reworded for desktop mode; on the headset the facilitator spoke the controller versions above.
- P1 and P2's records were created on Thu 24 Sep (dry-run entries) and overwritten on the day; all values shown were entered on 25 Sep.
- Observer notes in §2 are lightly tidied for spelling; the raw text is: P1 "different interaction with the lamp (fysical interaction) · Gaming interaction like ballon trough in the wall and change the colour"; P2 "Other types of room"; P4 "Too sensitive the fortitude moving · Swap for the menu" ("fortitude" read as an autocorrect of "furniture").

D Session and device notes

ITEM	NOTE
Hardware	Meta Quest 3S (borrowed), controllers. APK sideloaded over USB on the day. Performance 72 fps held (logcat VrApi FPS=72/72; log in testing-data/ip2a/device/, last minutes only).
Session times (logger)	P1 ≈ 14:37 · P2 ≈ 14:59 · P3 ≈ 15:21 · P4 ≈ 15:40 · P5 ≈ 15:50.
Interruptions	Meta button pressed twice ≈ 15:19, during P3's session (system-reserved, can't be blocked). Headset slept 15:31–15:37 between P3 and P4 (proximity sensor); disabled from 15:38, so P4 and P5 ran without it.
Fix on the day, before sessions	Lamp, clock and preset frames only responded to physical touch on the headset; point + trigger was added so they can also be pressed with the ray (commit edd1d07).
Preset at start	Scandi living room for everyone, so the kept floor and sofa were identical across participants. App restarted between participants.